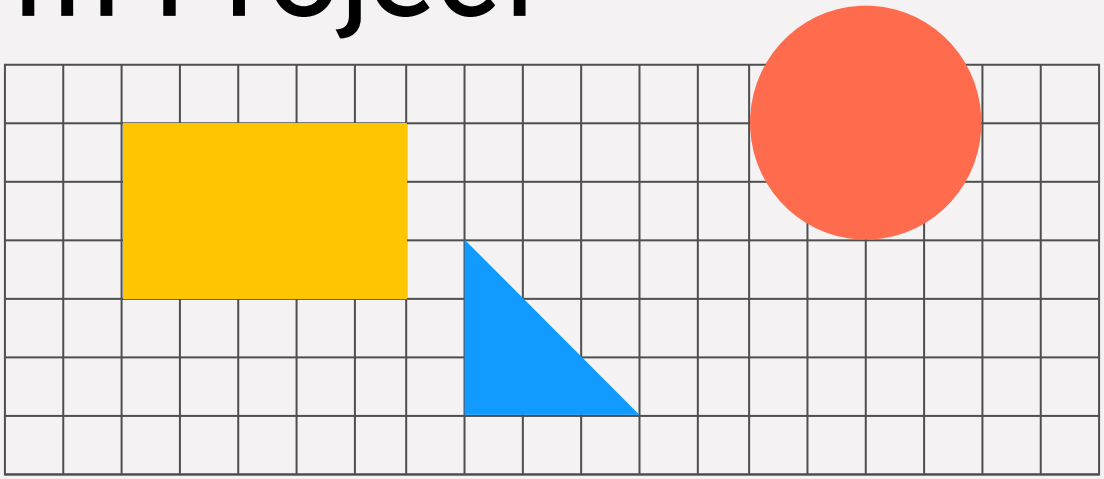
DSS 445: XLST R Statistical Programming Midterm Project

Visualizing and Manipulating Incomplete Datasets in R with the ‘naniar’ Package

Jacob Yan and Richard Le



OUR DATASET

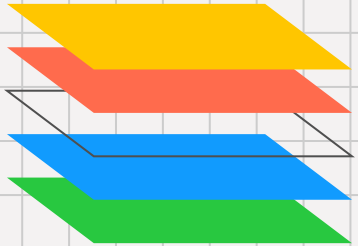
NHANES

The National Health and Nutrition Examination Survey (2009-'10, 2011-'12). It is a data frame consisting of 10,000 rows and 77 variables. It contains variables for participants' health with categories in body measurements, drug use, and other general health benchmarks. We used this dataset due to the large amount of missing data so that naniar could be used well.

```
> print(NHANES)
      ID SurveyYr    Sex SexOrientation
1  51624  2009_10   male   Heterosexual
2  51624  2009_10   male   Heterosexual
3  51624  2009_10   male   Heterosexual
4  51625  2009_10   male           <NA>
5  51630  2009_10 female   Heterosexual
6  51638  2009_10   male           <NA>
7  51646  2009_10   male           <NA>
8  51647  2009_10 female       Bisexual
9  51647  2009_10 female       Bisexual
10 51647  2009_10 female       Bisexual
11 51654  2009_10   male           <NA>
12 51656  2009_10   male   Heterosexual
```

What is ‘naniar’?





- Initially developed and published in 2016
- Team effort, led and primarily maintained by Nicholas Tierney
- Originally named ‘ggmissing’, but renamed over continued development
  - Distancing from ‘ggplot2’
- Triple-layered name
  - -Anagram of Narnia
  - -’NA in R [na-ni-aR]’, ‘naniar’ -> ‘nuhn-i-yahr’

Patches

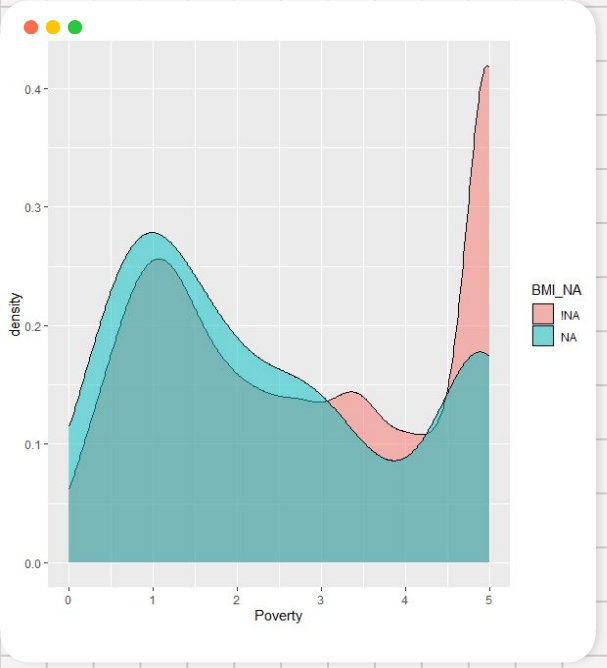
Many of naniar’s patches have referential names. Narnia continues to be visible, with the most recent patch ‘Prince Caspian’.

What can naniar do?

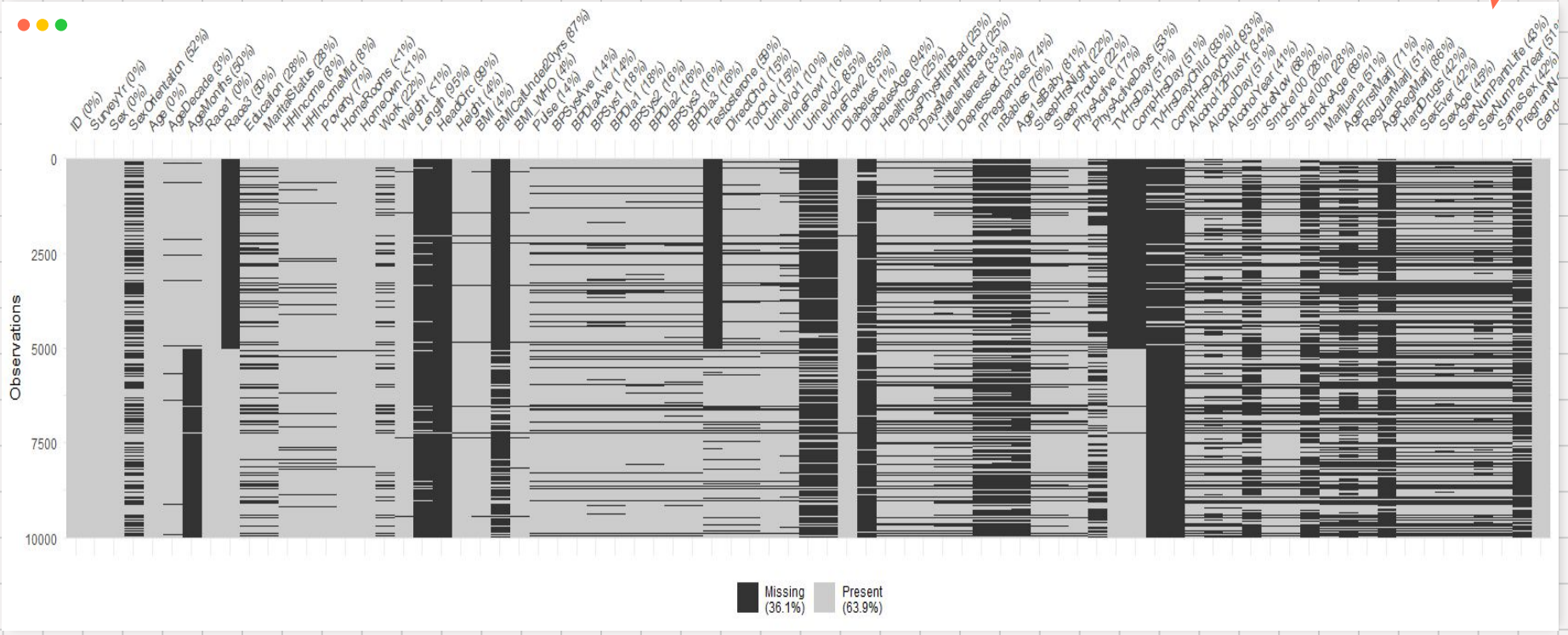
Hypothesis

We explored relationships in missing data. Our hypothesis was that Missing BMI was related to Poverty Level in the dataset.

- Can return the scale and concentration of missing values in a dataset
- Through visualizations or numerical summaries
- Create a data structure showing the absence or presence of data values
- Independently impute and plot imputed [formerly missing] values

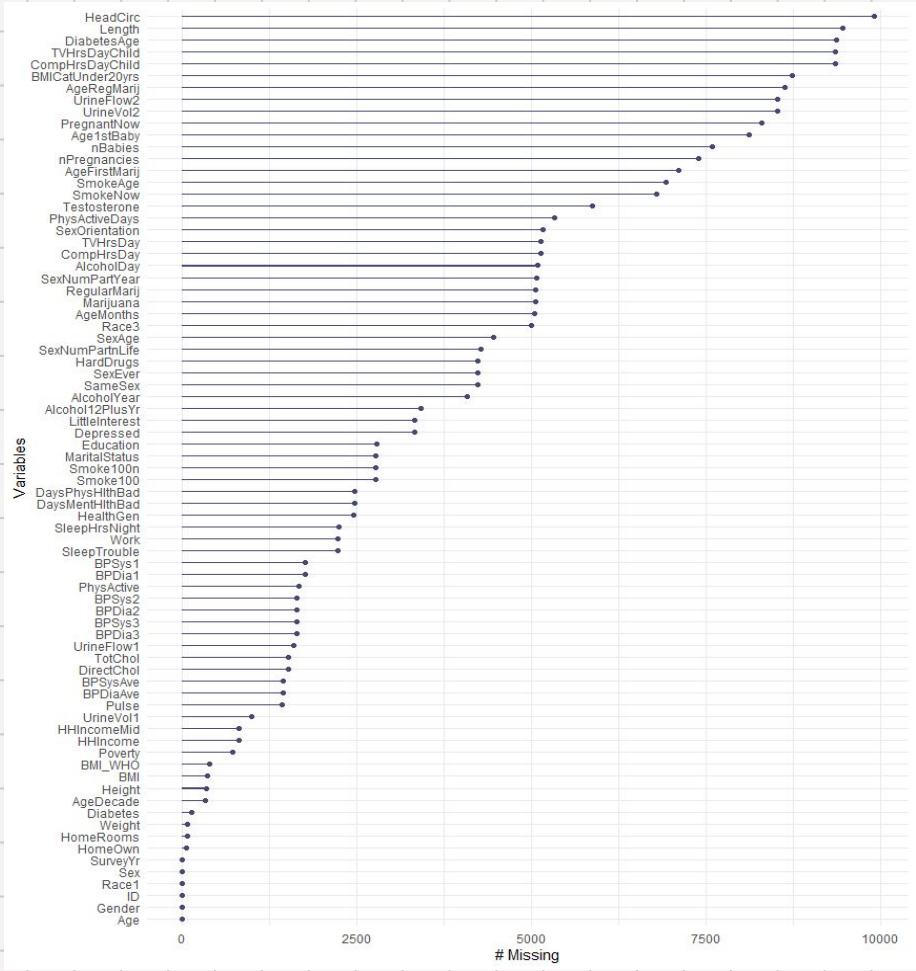


Visualizing with vis_miss()

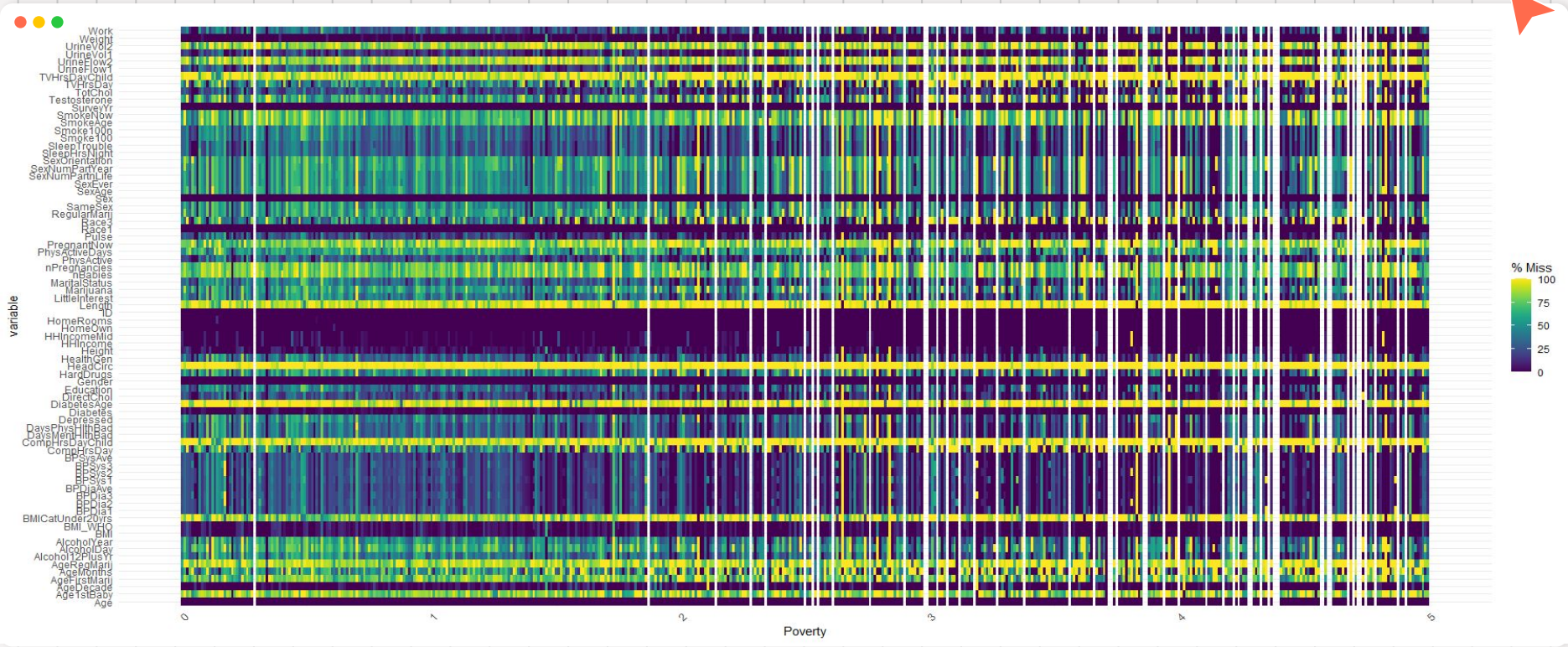


Visualizing with

gg_miss_var()



Visualizing w/ gg_miss_fct()



as_shadow & bind_shadow

```
> as_shadow(NHANES)
# A tibble: 10,000 × 77
   ID_NA SurveyYr_NA Sex_NA SexOrientation_NA
   <fct> <fct>       <fct>  <fct>
 1 !NA   !NA         !NA    !NA
 2 !NA   !NA         !NA    !NA
 3 !NA   !NA         !NA    !NA
 4 !NA   !NA         !NA    NA
 5 !NA   !NA         !NA    !NA
 6 !NA   !NA         !NA    NA
 7 !NA   !NA         !NA    NA
 8 !NA   !NA         !NA    !NA
 9 !NA   !NA         !NA    !NA
10 !NA   !NA         !NA    !NA
```

